CR-04 Requirements Specification Writing Guide

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1. Document metadata

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Document ID	CR-04
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Category	A - Client requirements
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Data status	Factual guidance with synthetic examples drawn from the fictional jobs
Main topics	Requirements writing; Hard versus soft constraints; Requirement IDs
Relevant skills and technologies	Requirements analysis
Relevant entity IDs	JOB-001, JOB-002, JOB-003, JOB-005, CR-03, EVA-005

Purpose and scope: explain how client requirements are written, numbered and classified so freelancers can estimate them and an evaluator can verify them.

2. Executive summary

Good requirements are numbered, measurable and classified as mandatory or preferred. The guide shows weak and strong examples from the fictional jobs and a specification template.

3. Purpose of a requirements specification

A requirements specification states what the client needs in a form that freelancers can price and that an evaluator can check. It turns a project brief into numbered, testable statements. In this knowledge base, CR-01 and CR-02 hold the briefs, CR-03 holds the specifications and acceptance criteria of JOB-001 to JOB-010, and this document explains how they are written.

4. Structure and ID scheme

ID pattern	Meaning	Example
JOB-NNN	A client job and its brief	JOB-001
REQ-NNN-NN	Functional requirement of job NNN	REQ-001-03
NFR-NNN-NN	Non-functional requirement of job NNN	NFR-001-01
AC-NNN-NN	Acceptance criterion of job NNN	AC-005-02

Stable IDs let a proposal and a retrieval chunk refer to exactly one requirement. Repeat the job ID in every section so a chunk keeps its meaning when extracted alone.

5. Writing measurable requirements

- State one observable behaviour per requirement.
- Give a number, a condition or a test for anything qualitative.
- Mark targets that are only examples as illustrative.
- Use the same skill names as the skill taxonomy (EK-08).
- State who provides inputs and by when.

Weak statement	Strong statement (from the dataset)
The order list must be fast.	Order list loads within 1 second for 20,000 orders (NFR-001-01, illustrative target).
The app should work offline.	Airplane-mode test shows booked classes (AC-005-02).
The dashboard should be accessible.	All controls are reachable by keyboard (REQ-002-04).
Answers must be good.	Every answer in a 20-question acceptance test includes at least one citation (AC-003-02).

6. Mandatory constraints versus preferences

Question	If yes	If no
Would the client reject a proposal that violates it?	Hard constraint (mandatory)	Preference
Is there an external date or limit (event, contract, legal)?	Hard deadline or hard budget limit	Soft deadline or preferred budget
Is the skill needed from day one with no learning time?	Mandatory skill (direct evidence required)	Required or optional skill

Hard constraints make a proposal ineligible when violated; preferences only lower a score (EVA-005). A client interface should force an explicit choice for budget, deadline and each skill.

7. Mapping requirements to skills in the dataset

Job	Mandatory skills	Other required skills	Optional skills
JOB-001	Angular, Spring Boot	PostgreSQL	Docker, Unit testing
JOB-002	Angular	TypeScript, RxJS	NgRx, Accessibility
JOB-003	Python, RAG pipeline	Vector search, FastAPI	LLM evaluation, Docker
JOB-004	Docker, Kubernetes	CI/CD, Terraform	Monitoring, AWS
JOB-005	Flutter	Dart	Offline storage, Firebase push, Unit testing
JOB-006	Python, scikit-learn	FastAPI	PyTorch, Spring Boot
JOB-007	Spring Boot, Angular	PostgreSQL, REST API design	Docker, Unit testing, CI/CD
JOB-008	Spring Boot, PostgreSQL	none	Redis, Unit testing
JOB-009	Angular	TypeScript, Unit testing, Accessibility	Spring Boot
JOB-010	Docker, CI/CD	none	Security scanning, Monitoring

8. Specification template

- Job ID, title, client, posting date.
- Business context and goals.
- In-scope and out-of-scope lists.
- Functional requirements with IDs and priority (Must or Should).
- Non-functional requirements with measurable targets.
- Acceptance criteria linked to requirement IDs.
- Constraints: budget (preferred and hard), deadline (date and type), mandatory skills.
- Dependencies and client responsibilities.
- Change control reference (EST-006).

9. Common mistakes

- Mixing a hard limit and a preference in one number.
- Omitting out-of-scope items, which later become disputed exclusions.
- Using unmeasurable words such as 'fast', 'modern' or 'intuitive'.
- Listing every technology as mandatory.
- Leaving dependencies undated.

10. Evidence requirements and uncertainty

When a requirement is ambiguous the assistant must say which requirement ID is unclear and must not invent a target. If a job has no stated deadline type, the deadline is treated as soft and the explanation says so.

11. Summary of key findings

- Use stable IDs and measurable targets.
- Separate hard constraints from preferences.
- Out-of-scope lists prevent later disputes.

12. Keywords and alternative terminology

requirements specification; SRS; requirement writing; SMART requirements; mandatory constraint; hard limit; preference; template